THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL

FORM FOUR PRE-MOCK EXAMINATION
031 PHYSICS
MARKING SCHEME 2025

SECTION A (16 MARKS)

1. Answer……10 marks@1

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	D	B	C	A	A	D	C	D	D

2. ……6 marks @1

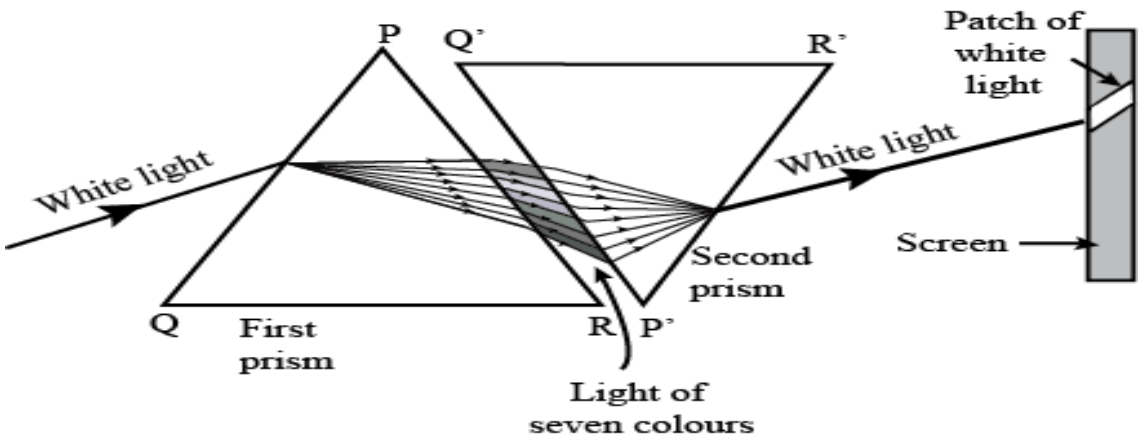
List A	i	ii	iii	iv	v	vi
List B	F	H	G	E	B	C

SECTION A (54 MARKS)

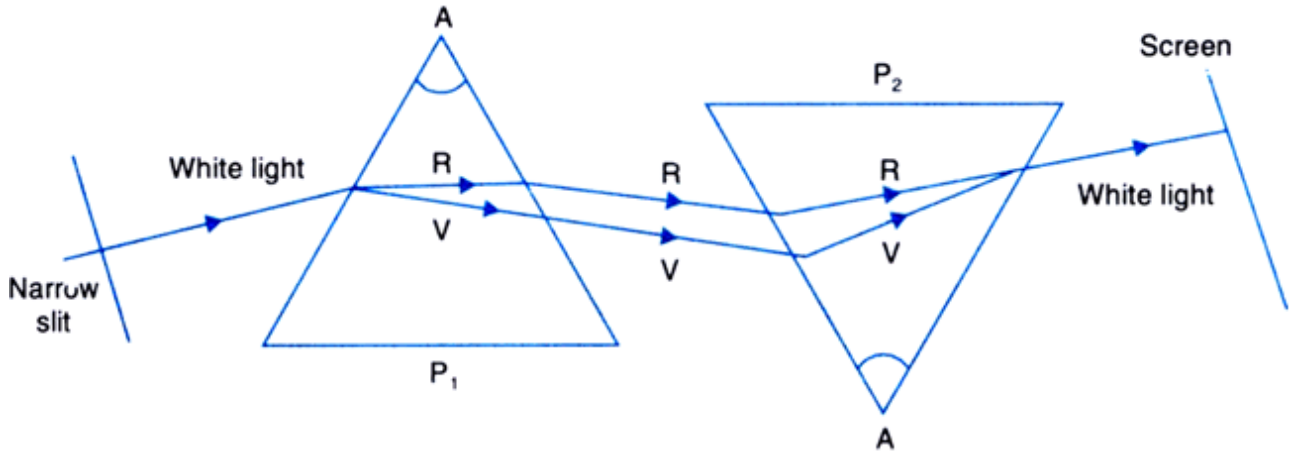
3.

(a) Recombination of dispersed white light ……4 marks

– Occurs when the second prism is reversed with respect to first prism



OR

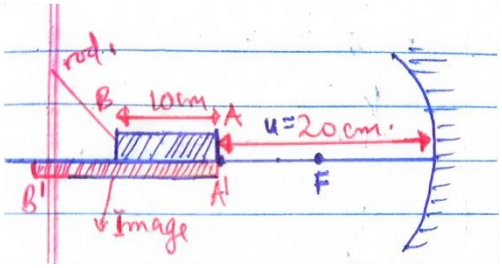


(b) Using mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

Sketch



For the first end of the rod (end A)

$\frac{1}{V_1} = \frac{1}{f} - \frac{1}{u}$

$V_1 = \frac{fu}{u-f}$

$v_1 = \frac{10 \times 20}{20-10}$

$v_{1=20cm}$

For the second end u become 20cm +10cm

U = 30cm

Then $v_2 = \frac{fu}{u-f}$

$v_2 = \frac{10 \times 30}{30-10}$

$v_2 = 15cm$

Then the length of the image = 20 cm -15 cm

= 5 cm. ……..5 marks

4. (a)(i)

✓ Because pressure is greater at the bottom than at the surface……..02 marks

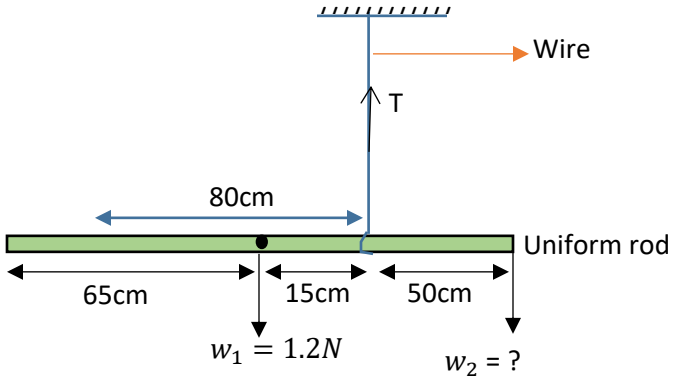
(ii)

Many tyres increases surface in contact with the ground which reduce average pressure of truck. Hence the truck does not tear the road or stuck in the ground ……02 marks

(b) Solution

(i) Rods weight w_1=mg = 0.12 x 10 = 1.2N (at the centre of the rod)

Let w_2 hang from the right end



Weigh of the rod (w_2) cause anticlockwise moment while unknown weight (w_2) cause clockwise moment

From principle of moments

$$\sum c.m = \sum a.c.m$$

$$w_1 d_1 = w_2 d_2$$

$$w_2 = \frac{w_1 d_1}{d_2}$$

$w_2 = \frac{1.2N \times 0.15m}{0.5m}$ = 0.36N = 36g…….3marks

Therefore mass of 36g is required to be hang to the right end.

(ii) For rod to be in equilibrium , tension must be equal to the total weight

T = 1.2N + 0.36N

T = 1.56N……2 marks

5. (a) (i)

- If a body is moving with a constant velocity, there is no external force acting on it, therefore its acceleration is zero.

F = ma

But a = 0m/s^2.

Then F = m x 0

F = 0N……..02marks

(ii)

- No , because there may be some balanced forces acting on object ,hence object not move ……..2marks

(b)

(i) VR = $\frac{number\ of\ teeth\ on\ drivven\ gear}{number\ of\ teeth\ in\ driving\ gear} = \frac{80}{20}$

V.R = 4……02marks

(ii) Let

R_A = Rotation in A

T_A = Number of teeth in A

R_B = Number of rotation in B

T_B = Number of teeth in B

$R_A \times T_A = R_B \times T_B$

$3 \times 20 = R_B \times 80$

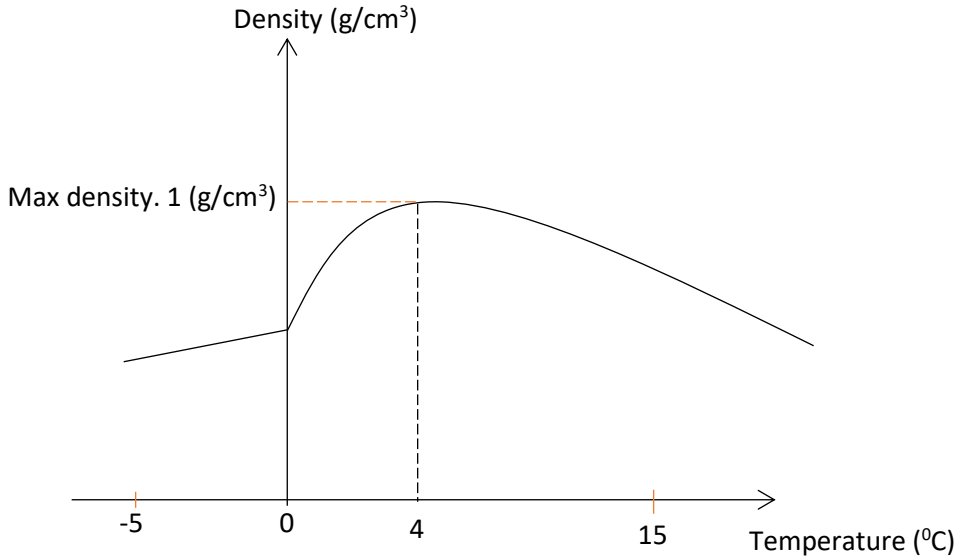
$R_B = \frac{60}{80} = \frac{3}{4}$……..03marks

6. (a) A beaker containing water ice is heated from -5^{0}C to 0^{0}C and then from 0^{0}C to 15^{0}C. With the aid of a diagram, explain the variation of density with temperature. (4 marks)

Answer

When the ice is heated at -5^{0}C 0^{0}C then heated to 15^{0}C, its density will increase from the temperature of -5^{0}C to 4^{0}C where it is maximum with value of 1g/cm^3.Heating above 4^{0}C the density of the liquid decreases

Graph of density against temperature.



(b) A heater of 500w is used to raise the temperature of 50kg of material of specific heat capacity of 960J/kgK, from 18^{0}C to 38^{0}C. Assume that all the heat from the heater is given to the material. Calculate:

(i) Heat capacity of the material **(**2 marks**)**

(ii) The time taken in seconds **(**3 marks)

Solution

(i) Heat capacity = mass x specific heat capacity

= 960 x 50

= 48000J/K……..02 marks

(ii) Heat supplied = mcΔT

= 960 X 50 (38-18)

= 960,000J…….03 marks

But

Heat = Power x time

Time, t = $\frac{heat\ supplied}{power} = \frac{960{,}000}{500}$ = 1920seconds.

7. (a) (i) The half-life of a certain radioactive substance is 64 days. Explain the meaning of this statement.**(2 marks)**

Answer

It means it takes 64 days for radioactive substance to decay to decay to its half

(ii) The half-life of 60_{C_o} is 5 years .How much time does it take for $(\frac{3}{4})^{th}$ of its initial mass to disintegrate?(4 marks)

Solution

Initial atomic mass No = 60

$t_{\frac{1}{2}}$ = 5years

$time$ t =? Final amount N = ¾ X 60 = 15

From $\frac{N}{N_o} = \left(\frac{1}{2}\right)^{\frac{t}{5}}$

$\frac{15}{60} = \left(\frac{1}{2}\right)^{\frac{t}{5}}$

$\frac{1}{4} = \left(\frac{1}{2}\right)^{\frac{t}{5}}$

$\left(\frac{1}{2}\right)^{2} = \left(\frac{1}{2}\right)^{\frac{t}{5}}$

$2 = \frac{t}{5}$

t = 10 years ……...04 marks

(b) Thorium disintegrates in the following manners

$90^{Th^{232}}.\rightarrow 88^{Ra^{228}} \rightarrow 89^{Ac^{228}}.\rightarrow 90^{Th^{228}} \rightarrow 88^{Ra^{224}}$.State the particle emitted at each part of the disintegration. (4 marks)

Answer

$90^{Th^{232}}.\rightarrow 88^{Ra^{228}} + {}^{4}_{2}\alpha$ $(\alpha - particle)$……..01mark

$88^{Ra^{228}} \rightarrow 89^{Ac^{228}} + {}^{0}_{-1}\beta$ (β- particle) ……...01mark

$89^{Ac^{228}}.\rightarrow 90^{Th^{228}} + {}^{0}_{-1}\beta$ (β- particle)……...01mark

$90^{Th^{228}} \rightarrow 88^{Ra^{224}}.+ {}^{4}_{2}\alpha$ $(\alpha - particle)$……...01mark

$\therefore\ 90^{Th^{232}}.\xrightarrow{\alpha} 88^{Ra^{228}}.\xrightarrow{\beta} 89^{Ac^{228}}..\xrightarrow{\beta} 90^{Th^{228}}.\xrightarrow{\alpha} 88^{Ra^{224}}$

8. (a) Heliocentric theory is a theory which puts forward that the sun is at the centre of the solar system and all other bodies including the earth revolve around it…….2 marks

Shortfall of heliocentric theory ……..3 marks @ 01mark

- The sun was thought to be at the Centre of the universe but it is only the Centre of the solar system.
- Orbits of the planets were thought to have circular shapes , but now days they are known to have elliptical shapes
- The relative positions of the stars seemed to remain the same despite the earths changing view points as it moved around the sun.

(b)

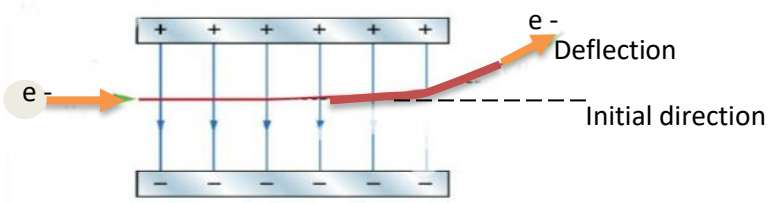
(i) A – Troposphere
Temperature in troposphere decrease because gases in this layer do not absorb radiation that enters the atmosphere……..1 mark

(ii) Temperature in layer B increase due to presence of ozone layer that absorbs ultraviolet radiation.
The layer B is stratosphere ……..1 mark

(iii) Because it consist of charged particle called ions and free electrons responsible for propagation of electromagnetic wave. Radio waves can be reflected off the ionosphere allowing radio communication over long distance ……..2 mark

SECTION B (30 MARKS)
Answer only two (2) questions from this section

9. (a) (i) The electron will be deflected towards the positive plate as seen in figure below.



(ii)

- Walking barefooted during lightning storm will facilitate large amount of current from lighting strike to pass through your to the ground causing severe damage on you or some time cause death.

(b)

Solution

Hints R = $2R_0$ V = $\frac{1}{3}V_0$

I = $\frac{V}{R} = \frac{\frac{1}{3}V_0}{2R_0} = \frac{V_0}{6R_0}$ but $V_0 = I_0R_0$

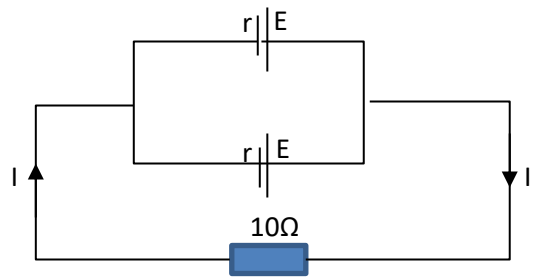
I = $\frac{I_0R_0}{6R_0}$

$\frac{I}{I_0} = \frac{1}{6}$

(c)

Solution

Given two dry cells each with emf,E = 1.5V
Internal resistance, r = 1.0Ω
External resistance, R = 10Ω
Since the cells are in parallel;
Total emf = 1.5V



Equivalent resistance = $R + \frac{r \times r}{r + r}$

$= R + \frac{r}{2}$

$= 10 + \frac{1}{2}$

$= 10.5\Omega$

Current, $I = \frac{V}{R} = \frac{1.5V}{10.5\Omega} = 0.143A$

Power = I^2R

$= 0.143^2 \times 10$

= 0.20W,……5 marks

10. (a) A strong current when passes through a semiconductor heats up the semiconductor and the covalent bonds break up. It results in a large number of free electrons. The material then behaves just as a conductor. As now the semiconductor no longer possesses the property of low conduction, it is said to be damaged……4 marks

(b)

(i) The statement means the small change in collector current cause greater change in base current……..2 marks

(ii) $I_C \propto I_B$

$I_C = \beta I_B$

Where β = current gain

So

$I_C = \beta I_B$……3 marks

(iii) $I_B = 2\mu A$

I_C =?

$\beta = 200$

Then

$I_C = \beta I_B$

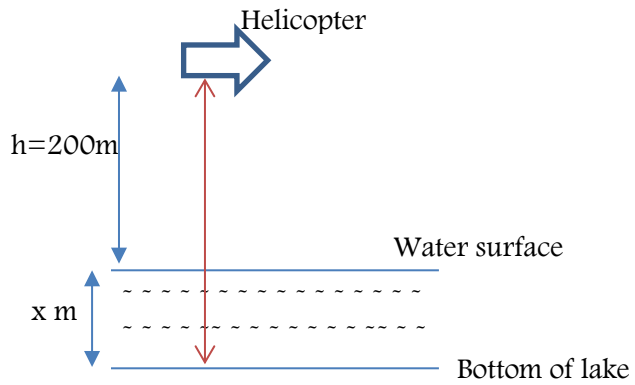
$I_C = 200 \times 2 \times 10^{-6}$

$I_C = 4 \times 10^{-4} A$

$I_C = 400\mu A$. ……3 marks

(b) When electrons are ejected from the negatively-charged electroscope, they escape, but in a positively-charged electroscope they are attracted back by the positive charge on the plate……3 marks

11. (a) Nature of the problem ……..01 mark



From $v = \frac{2d}{t} \rightarrow t = \frac{2d}{v} = \frac{2d}{331.4+0.6\theta}$……0.5mark

Let t_1 = time taken by sound to reflect back from water surface,

$t_1 = \frac{400m}{331.4+0.6\theta}$……..0.5mark

Let t_2 = time taken by sound to reflect back from bottom of lake,

$t_2 = \frac{2(200m+xm)}{331.4+0.6\theta}$……0.5mark

Difference in arrival time = 0.24s

$t_2 - t_1$ = 0.24s

$\frac{400+2x}{331.4+0.6\times20} - \frac{200}{331.4+0.6\times20}$ = 0.24sec……..0.5marks

400 + 2x

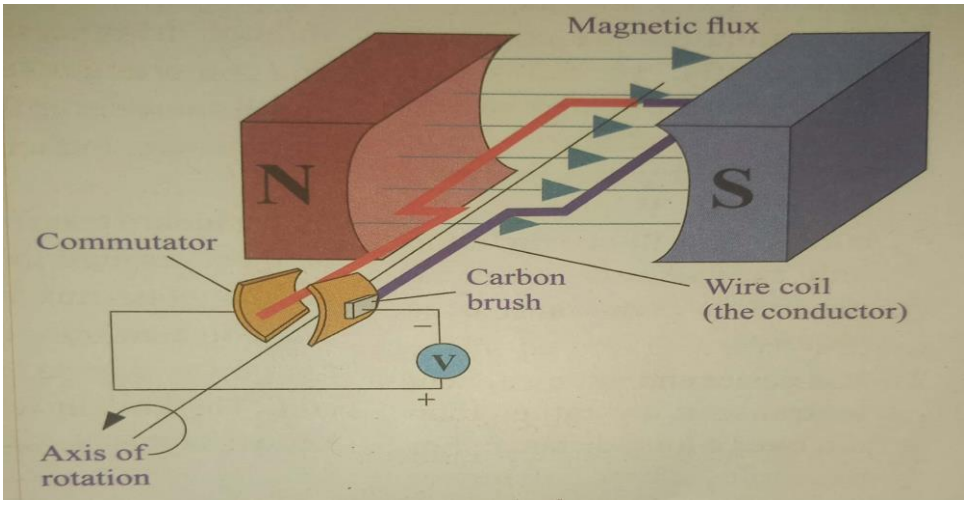
400 +2x = 480.76

2x = 80.76

x = 40.38

The depth of the lake = 40.38m…….03marks

(b) (i) diagram ……4marks



(b) A coil rotates because as the current flows through it, a magnetic field is developed around the coil. This magnetic field is always such that it opposes the existing magnetic field resulting into repulsion (Lenz's law); therefore, a force is exerted on the coil since the coil is free to rotate then it does so;
The direction of rotation is perpendicular to the plane made by current and direction of the field. (Fleming's left hand rule)……..3 marks

(c) When the coil is perpendicular to the fields, it will not be able to rotate unless it is slightly

pushed;…….03 marks